

# CE 310

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## The Poisson Distribution

Week 4 · Session 1 · When Rare Events Have a Rate

University of Arizona · Dr. Wooyoung Jung · Fall 2026

# Part 1

When the Label Lies

# The Drawing Needs Your Seal

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You are the engineer of record for a **detention basin in Columbia, South Carolina**. The hydrology report sizes it to the design storm required by the local standard. Everything is in order and the drawing is ready for your seal.

A junior engineer stops you with one page from the rainfall record:

- **October 4, 2015** — 27 inches in 48 hours, declared a "**1,000-year flood**"
- Two **comparable events** in the previous six years

**Three "1,000-year" floods in six years.**

 You cannot commission a new study — the record is what it is. **Do you seal the drawing?**

## If the 1,000-Year Label Were Correct...

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A "1,000-year" event has  $1/1000 = 0.001$  probability per year. Over **6 years**, the expected count is:

$$\lambda = 6 \text{ years} \times 0.001 / \text{year} = \mathbf{0.006} \text{ events in 6 years}$$

Under this model — accepting the label at face value — the probabilities are:

| Event                            | Probability         |
|----------------------------------|---------------------|
| $P(X = 0 \text{ in 6 years})$    | 99.40%              |
| $P(X \geq 1 \text{ in 6 years})$ | 0.60%               |
| $P(X \geq 3 \text{ in 6 years})$ | <b>&lt; 0.0002%</b> |

The model says: in a world where the label is correct, virtually nothing should happen in 6 years.

**But Columbia experienced three such events in six years.**

# What Three Events in Six Years Prove

Columbia observed three "1,000-year" events in six years. Two explanations:

| Explanation              | Assessment                                  |
|--------------------------|---------------------------------------------|
| Extraordinarily bad luck | < 1 in 500,000 — not a credible explanation |
| The label was wrong      | Far more likely                             |

**You send the drawing back — and you can now say why in one sentence:** the design storm rests on a return period the rainfall record contradicts at better than 500,000 to 1.

💡 The Poisson model cannot say *why* the label was wrong — bad data, short record, or shifting climate. But it says the label **is** wrong, with near-certainty. This is what quantitative models do: they make wrong assumptions visible.

# How a Wrong Label Gets Into the Standard

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No one has observed a "1,000-year flood" directly — the label comes from extrapolating a statistical model well beyond the available data.

## **The core problem:**

Columbia's gauge record spans ~80 years. Engineers fit a distribution to those observations, then extrapolated far into the tail to estimate the 0.1%-per-year flood level — and called it "1,000-year."

Projecting 12× beyond the record is like predicting a car will last 600,000 miles after driving it only 50,000.

## How a Wrong Label Gets Into the Standard (cont.)

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### Two compounding errors:

- **Distribution choice:** Different statistical fits to the same 80 years of data produce very different 1,000-year estimates. There is no uniquely correct answer.
- **Non-stationarity:** A calibration from historical data assumes conditions haven't changed. Urbanization and shifting climate can silently raise the true flood rate.

⚠ These are not errors of negligence — they are limitations of the method. The Poisson model does not fix them; it reveals when the resulting label contradicts what we actually observed.

# The Same Error Appears in Engineering

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| Design decision                    | Wrong approach           | Consequence                                 |
|------------------------------------|--------------------------|---------------------------------------------|
| Size a roof drain                  | Annual average rain rate | Undersized for monsoon bursts               |
| Specify HVAC maintenance intervals | "Replace when it breaks" | Unplanned failures during peak season       |
| Declare a storm return period      | One data point, no model | Infrastructure fails at "impossible" events |
| Estimate bridge fatigue life       | Mean traffic load        | Actual failures exceed predictions          |

⚠ The common thread: using a **mean** or a **label** where a **distribution** is required. The average conceals the tail. The Poisson distribution makes the tail explicit.

# The Question We Need to Answer

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We know that rare events — rain days, equipment faults, extreme storms — occur at some **average rate** over time.

## What we need:

- Not just *how often on average*, but *how likely is  $k$  events in a given period?*
- Not just *the mean*, but *the full distribution of counts*

## What we're building today:

A probability model for **discrete counts of events** that occur at a known average rate — the **Poisson distribution**.

# Part 2

## The Poisson Model

# Why We Need a Distribution

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The mean ( $\lambda = 3.86$  rain days/month for Tucson) tells you what is *typical*. It does not answer engineering questions.

| Question                       | Mean alone | Full distribution |
|--------------------------------|------------|-------------------|
| Expected rain days this month? | ✓ 3.86     | ✓                 |
| P(exactly 8 rain days)?        | ✗          | ✓                 |
| P(more than 6 rain days)?      | ✗          | ✓                 |
| P(dry month — zero rain days)? | ✗          | ✓                 |

The **Probability Mass Function (PMF)** assigns a probability to every possible integer count  $k$ :

$$\text{PMF: } P(X = k) \quad \text{for } k = 0, 1, 2, \dots$$

For Poisson, the PMF is determined entirely by one parameter:  $\lambda$ . Every question in the right column is answered by evaluating the PMF at the appropriate  $k$ .

# Poisson's Original Problem — and Why It Works

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**Siméon Denis Poisson, 1837:** Original goal was modeling judicial errors — how often do juries wrongfully convict? He needed to count rare, independent events across a large number of trials.

**The mathematical key** — Poisson is the limit of the Binomial:

$$n \rightarrow \infty, \quad p \rightarrow 0, \quad n \times p = \lambda \text{ (constant)}$$

When any single event is rare but you have many chances for it, the Binomial simplifies to a one-parameter model. One number —  $\lambda$  — carries the entire distribution.

**First engineering use:** In 1898, Ladislaus Bortkiewicz counted deaths by horse kick in the Prussian cavalry — 14 corps over 20 years. Rare events, large exposure, constant average rate. The Poisson fit was near-perfect. The same three conditions now define every CE application you will use this week.

# Three Conditions for Poisson to Apply

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The Poisson distribution is appropriate when:

1. **You are counting events** — discrete occurrences in a fixed time or space (rain days per month, faults per year, calls per hour)
2. **Events are independent** — whether or not event  $k$  occurred does not affect the probability of event  $k + 1$
3. **Average rate  $\lambda$  is constant** over the observation period

⚠ Poisson breaks down when events cluster (aftershocks following an earthquake), or when the rate changes systematically (monsoon season vs. winter). The solution: **condition first, then apply Poisson**. Seasonal  $\lambda$  values (July vs. Spring) handle the varying-rate problem exactly this way.

# The PMF Formula

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$$P(X = k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!} \quad k = 0, 1, 2, \dots$$

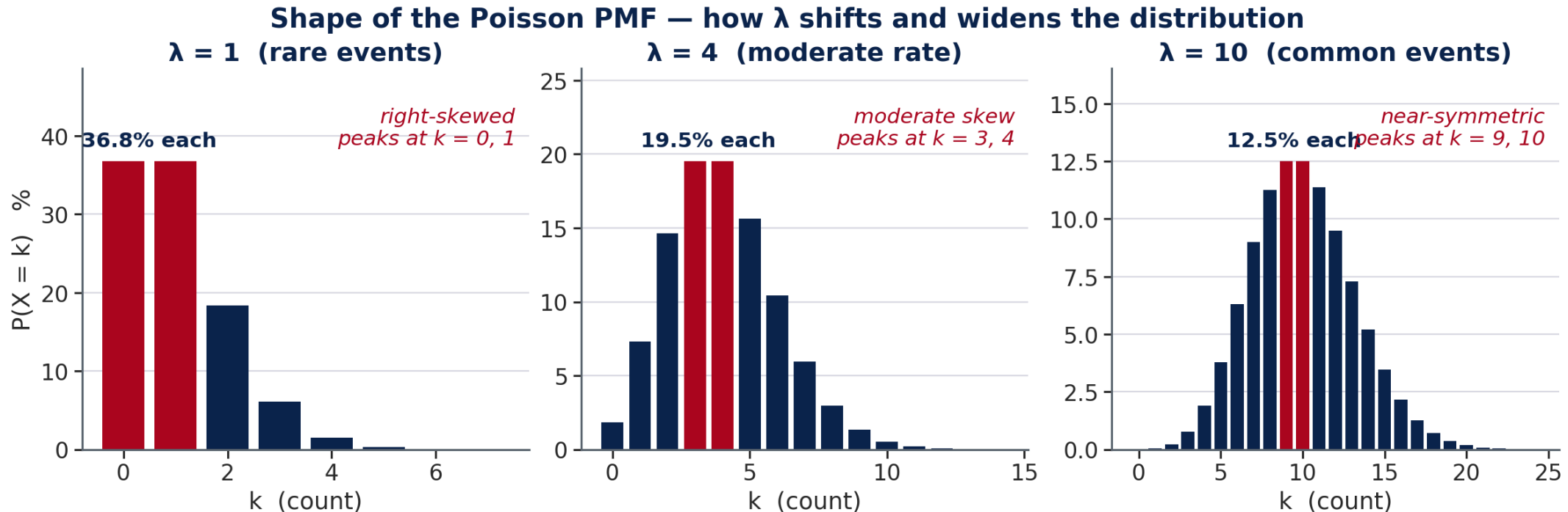
Each piece:

| Symbol           | Meaning                                      |
|------------------|----------------------------------------------|
| $k$              | The count you're asking about (0, 1, 2, ...) |
| $\lambda$        | The average rate (events per period)         |
| $e^{-\lambda}$   | Probability of zero events — the "baseline"  |
| $\lambda^k / k!$ | Relative weight of exactly $k$ events        |

You do not need to compute this by hand in Excel — `=POISSON.DIST(k, λ, FALSE)` does it directly. But understanding the formula tells you *why* the distribution has the shape it does.

# Shape of the PMF: How $\lambda$ Changes Everything

**Why the shape changes:** At small  $\lambda$ ,  $P(X = 0) = e^{-\lambda}$  is large — zero events is the most probable outcome, creating strong right-skew. As  $\lambda$  grows, zero becomes rare, the mode shifts toward  $\lambda$ , and the distribution symmetrizes. Skewness =  $1/\sqrt{\lambda}$  — it shrinks toward zero as  $\lambda$  grows, approaching the bell shape of the Normal distribution (Week 5).



# What $\lambda$ Tells You — Mean and Variance in One Number

For a Poisson distribution,  $\lambda$  is both the **mean** and the **variance**:

$$E[X] = \lambda \quad \text{Var}[X] = \lambda \quad \text{SD}[X] = \sqrt{\lambda}$$

💡 **Why equal?** Each event is an independent rare occurrence — the count's mean and variance both trace back to the same rate  $\lambda$ . Compare the Normal distribution (Week 5), where mean and variance are separate free parameters. In Poisson, one number controls everything.

**Tucson overall:  $\lambda = 3.86$  rain days/month**

- Mean: 3.86 rain days
- Std deviation:  $\sqrt{3.86} \approx \mathbf{1.96}$  rain days
- A month with 6 rain days is  $\approx 1$  SD above average

**July monsoon:  $\lambda = 10.3$  rain days/July**

- Mean: 10.3 rain days
- Std deviation:  $\sqrt{10.3} \approx \mathbf{3.21}$  rain days
- A July with 17 rain days is  $\approx 2$  SD above average

# Answering the Week 3 Teaser

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At the end of Week 3 we asked:

*"If extreme-cooling hours occur at  $\lambda = 4$  per week during July, what is  $P(\text{exactly 6 such hours in a week})$ ?"*

$$P(X = 6 \mid \lambda = 4) = \frac{e^{-4} \cdot 4^6}{6!} = \frac{0.01832 \times 4096}{720} \approx \mathbf{10.42\%}$$

In Excel: `=POISSON.DIST(6, 4, FALSE)` → **0.10420**

💡 10.42% — not negligible, not certain. The Poisson model says extreme-cooling weeks happen about one in ten times during July. An HVAC system designed only for the mean load ( $\lambda = 4$  hours/week) will be undersized in roughly 40% of July weeks.

## Quick Check — Part 2

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### Before we continue

**Q:** In a building with  $\lambda = 2$  equipment faults per quarter, what is  $P(\text{zero faults this quarter})$ ?

Use the formula:  $P(X = 0) = e^{-\lambda}$

## ✓ Quick Check — Answer

$$P(X = 0 \mid \lambda = 2) = e^{-2} \approx 13.5\%$$

$$P(X = 0 \mid \lambda = 2) = e^{-2} = 0.1353$$

| Approach                                       | Result            |
|------------------------------------------------|-------------------|
| Formula: $e^{-\lambda}$                        | $e^{-2} = 0.1353$ |
| Excel: <code>=POISSON.DIST(0, 2, FALSE)</code> | 0.1353            |

**What this means:** A building with  $\lambda = 2$  equipment faults per quarter has about a **1-in-7 chance** of a completely fault-free quarter. Most semesters, at least one fault occurs.

💡 Students often overestimate this probability — they expect something closer to 50%. The exponential decay  $e^{(-\lambda)}$  falls faster than intuition suggests.

# Part 3

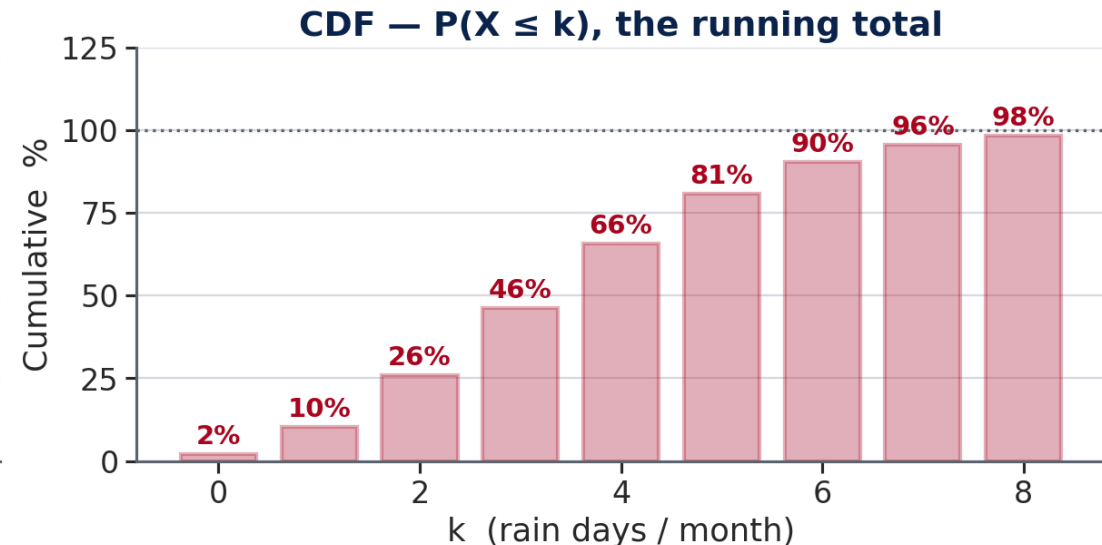
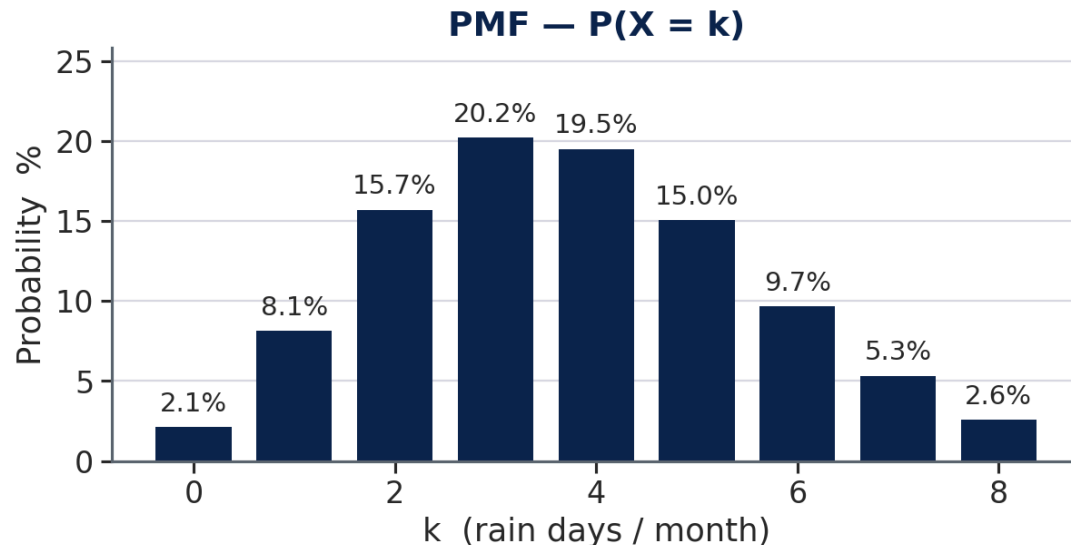
CDF and Exceedance Probability

# From PMF to CDF

The **PMF** answers:  $P(\text{exactly } k \text{ events})$ ? The **CDF** answers:  $P(k \text{ or fewer events})$  — the running total of PMF values from 0 up to  $k$ . → See Tutorial for the Excel step-by-step

| k | PMF   | CDF    |
|---|-------|--------|
| 0 | 2.10% | 2.10%  |
| 1 | 8.09% | 10.19% |

| k | PMF    | CDF    |
|---|--------|--------|
| 2 | 15.62% | 25.80% |
| 3 | 20.06% | 45.87% |



## Exceedance: $P(X > k)$

*How likely is the system to be stressed beyond design threshold  $k$ ?*

$$P(X > k) = 1 - P(X \leq k) = 1 - \text{CDF}(k)$$

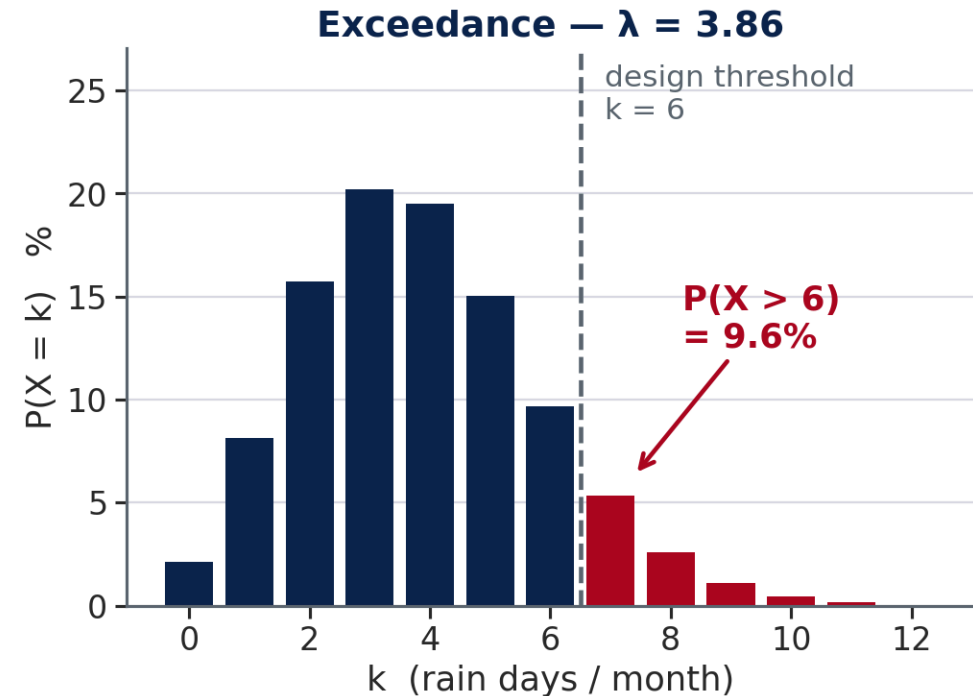
In Excel: `=1 - POISSON.DIST(k, λ, TRUE)`

**Example:**  $\lambda = 3.86$  — more than 6 rain days?

$$P(X > 6 \mid \lambda = 3.86) = 1 - 0.9034 = \mathbf{9.66\%}$$



One in ten months at the overall rate.  
Summer ( $\lambda = 6.57$ ) raises this to ~48% — why seasonal conditioning matters.



# Why Exceedance Is the Design Question

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| Design standard                                            | Exceedance framing                      |
|------------------------------------------------------------|-----------------------------------------|
| Size roof drain: no more than 5% of months exceed capacity | Find $k$ such that $P(X > k) = 5\%$     |
| HVAC filter: replace when 90% of months are below limit    | Find $k$ such that $CDF(k) = 90\%$      |
| Structural load: 2% in 50 years (ASCE 7 seismic)           | $P(X \geq 1 \text{ in 50 years}) = 2\%$ |

The question is never "what is the mean?" The question is always "what threshold ensures the tail probability is acceptable?"

# Part 4

Return Periods and Design Life

# What "Return Period" Actually Means

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A **return period**  $T$  years means the event occurs at an average rate of  $\lambda = 1/T$  **per year**.

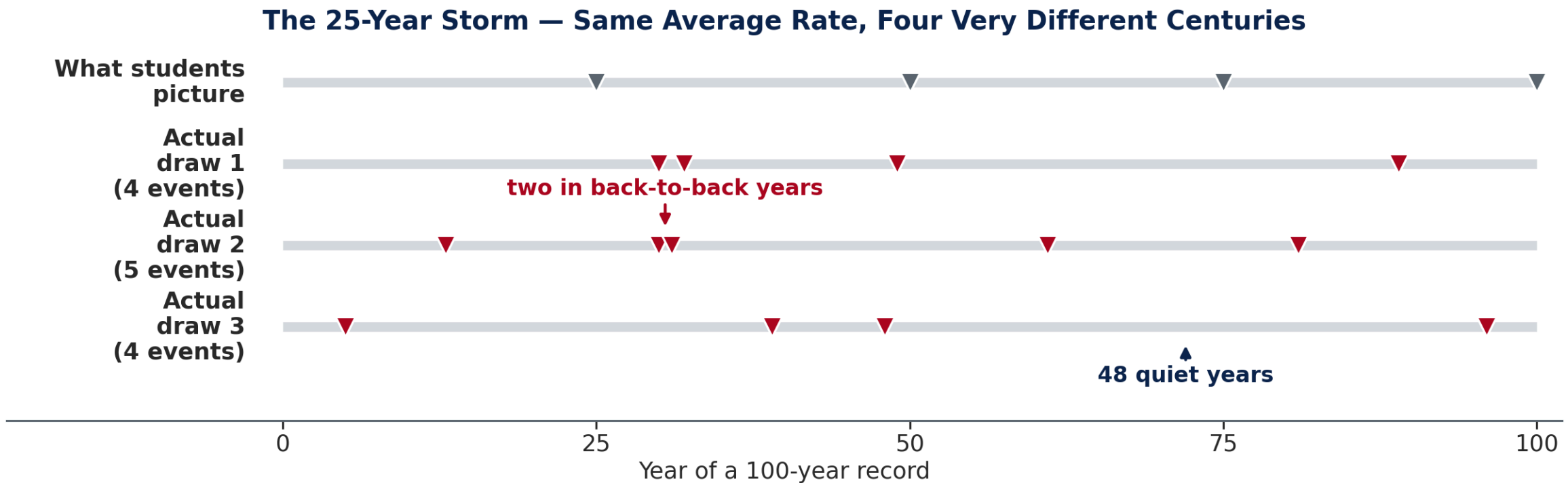
- **100-year storm:**  $\lambda = 0.01$  per year  $\rightarrow P(\text{occurs in any given year}) = 1\%$
- **50-year storm:**  $\lambda = 0.02$  per year  $\rightarrow P(\text{occurs in any given year}) = 2\%$
- **1,000-year flood:**  $\lambda = 0.001$  per year  $\rightarrow P(\text{occurs in any given year}) = 0.1\%$

**What return period does NOT mean:** "this event will occur exactly once every  $T$  years." Poisson events cluster randomly — two 100-year storms can occur in back-to-back years.

⚠ Engineers who say "the 50-year storm won't happen this decade because it already occurred last year" are confusing a return period with a schedule. The clock does not reset. Each year is independent.

# A Rate, Not a Schedule

Four centuries of the **same 25-year storm**. Only the top line is evenly spaced — and it is the one that never happens.



# Predict — The 100-Year Storm

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## 60 seconds · commit to a number before you compute

A box culvert is designed for the **100-year storm** — an event with a **1% chance in any given year**. The structure is expected to stay in service for a **50-year design life**.

**What is the probability it sees at least one 100-year storm in that time?**

- **A.** About 5%
- **B.** About 39%
- **C.** About 50%
- **D.** Essentially certain

*Write your letter down. Do not compute — commit to an intuition first.*

## ✓ 39.3% — Answer B

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$$P(\text{at least one}) = 1 - e^{-t/T} = 1 - e^{-50/100} = 1 - e^{-0.5} = \mathbf{39.3\%}$$

Not 50%. The annual chances **do not add** — that would double-count every year in which more than one storm occurs.

💡 Read it the other way and it is more unsettling: there is a **60.7%** chance the culvert never sees the storm it was designed for — and a nearly 40% chance it does. Neither number is what "one-in-a-hundred" sounds like.

# Probability Over a Design Life

The probability of at least one  $T$ -year event during an  $N$ -year project life:

$$P(\geq 1 \text{ event in } N \text{ years}) = 1 - e^{-N/T}$$

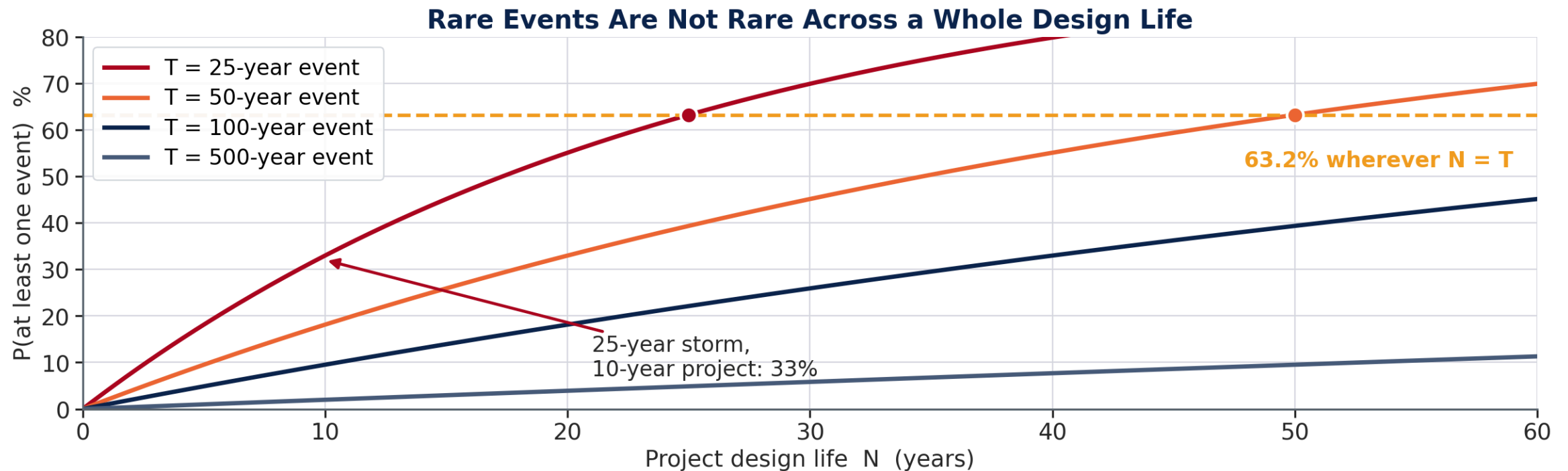
This is  $P(X \geq 1)$  under Poisson with  $\lambda = N/T$  events expected over the design life.

| Return period $T$ | 10-year project | 25-year project | 50-year project |
|-------------------|-----------------|-----------------|-----------------|
| 25-year storm     | <b>33.0%</b>    | 63.2%           | 86.5%           |
| 50-year storm     | 18.1%           | <b>39.3%</b>    | 63.2%           |
| 100-year storm    | 9.5%            | 22.1%           | <b>39.3%</b>    |
| 500-year storm    | 2.0%            | 4.9%            | 9.5%            |

**A 25-year storm has a 33% chance of occurring during a 10-year project — nearly one in three.**

# The Same Table, as Curves

Every curve is  $1 - e^{-N/T}$ . The gold line marks where project life equals return period — every curve crosses it at the same height.



# ASCE 7 and Design Return Periods

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ASCE 7-22 (Minimum Design Loads for Buildings and Other Structures) specifies structural design loads using explicit exceedance probabilities over a design life:

| Load type                     | Exceedance probability | Equivalent return period |
|-------------------------------|------------------------|--------------------------|
| Wind (Risk Category II)       | 7% in 50 years         | $\approx 700$ years      |
| Seismic (Risk Category II)    | 2% in 50 years         | $\approx 2,475$ years    |
| Flood (NFIP 1% annual chance) | 1% per year            | 100 years                |

→ Tutorial walks through the calculation in Excel

This is exactly the formula you will use in the in-class exercise cell B13 for a 25-year stormwater design event over a 10-year project life.

## Which Tool? — Counting vs. Magnitude

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Poisson answers: *How many* rain days occur in a month?

Here is a structurally different question: *How intense is the worst storm of the year?*

**For each row, which distribution family do you need?**

| Question                                     | Random variable       | Right tool |
|----------------------------------------------|-----------------------|------------|
| How many rain days this month?               | Count of events       | ?          |
| What is the peak 24-hour flow rate?          | Maximum of many draws | ?          |
| How strong is the worst wind gust this year? | Annual maximum        | ?          |

*Take thirty seconds. The middle column is the clue.*

## ✓ Poisson Counts Events — But Not How Bad They Are

| Question                                     | Random variable       | Right tool           |
|----------------------------------------------|-----------------------|----------------------|
| How many rain days this month?               | Count of events       | <b>Poisson</b>       |
| What is the peak 24-hour flow rate?          | Maximum of many draws | <b>Extreme value</b> |
| How strong is the worst wind gust this year? | Annual maximum        | <b>Extreme value</b> |

**Why it matters for design:** A drainage pipe fails when *peak flow* exceeds capacity — not when the count of rain days is high.

💡 Poisson models the *frequency* of events crossing a threshold — when the question shifts to *magnitude of the worst case*, a different distribution family applies.

# A Preview: Extreme Value Distributions

When the question shifts from *how many events* to *how extreme was the worst one*, a different model family applies.

| Poisson answers...             | Extreme Value answers...                   |
|--------------------------------|--------------------------------------------|
| How many rain days this month? | How intense was the worst storm this year? |
| Count of threshold crossings   | Magnitude of the annual maximum            |

ASCE 7 wind and flood loads come from fitting extreme value distributions to records of annual maxima — the return-period logic from Part 4 still applies, but the distribution family is different. The most common form is the **Gumbel distribution** (Type I Extreme Value), covered fully in Week 11.

 "How many" (Poisson) and "how extreme" (extreme value) are different questions requiring different tools. Week 11 goes deep.

# Part 5

Estimating  $\lambda$  from Data

# λ from Data: Count Events, Divide by Periods

$$\hat{\lambda} = \frac{\text{total events observed}}{\text{total observation periods}}$$

This is the **maximum likelihood estimator** for λ — the simplest and most natural estimate.

**Example — Tucson precipitation, 2013–2022:**

| Calculation                                  | Value                        |
|----------------------------------------------|------------------------------|
| Total rain days in dataset                   | 463                          |
| Total months in dataset                      | 120 (10 years × 12)          |
| $\hat{\lambda}_{\text{overall}} = 463 / 120$ | <b>3.858 rain days/month</b> |

💡 This single number — 3.858 — is both the mean and the variance of the monthly rain-day distribution.

# Conditioning Changes $\lambda$ , Not Just the Probability

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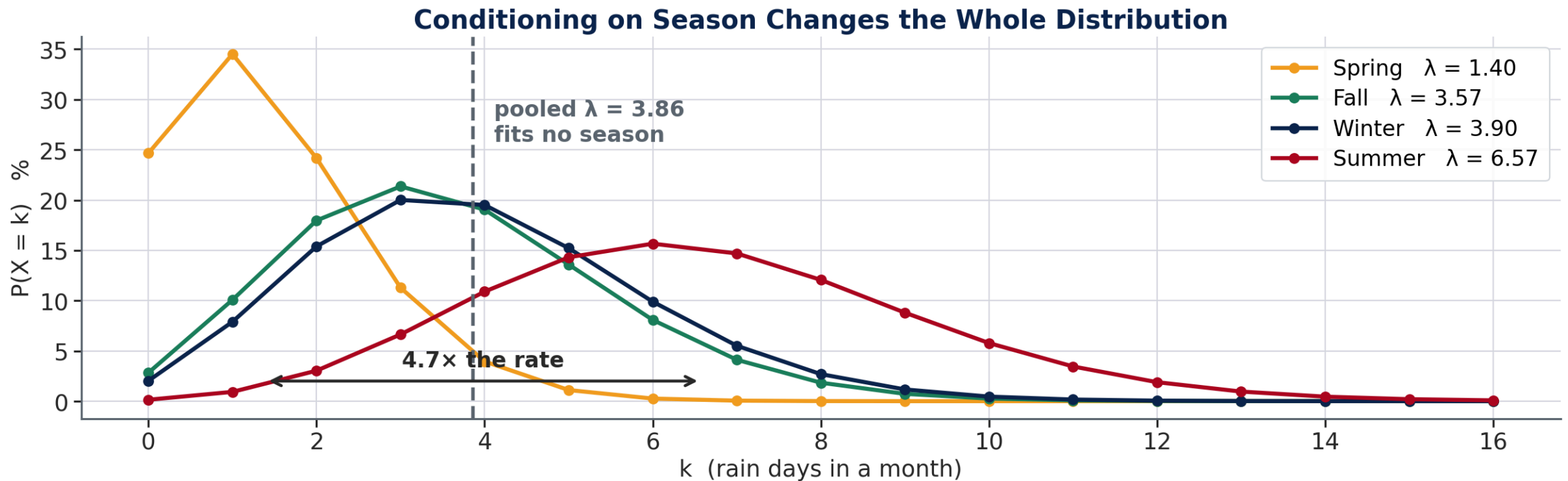
In Week 3 we conditioned on season to compute  $P(\text{event} \mid \text{context})$ . Now we see that conditioning on season changes the **entire distribution** — it changes  $\lambda$  itself.

| Season       | Rain days (2013–2022) | Months | $\hat{\lambda}$ |
|--------------|-----------------------|--------|-----------------|
| Winter (DJF) | 117                   | 30     | <b>3.90</b>     |
| Spring (MAM) | 42                    | 30     | <b>1.40</b>     |
| Summer (JJA) | 197                   | 30     | <b>6.57</b>     |
| Fall (SON)   | 107                   | 30     | <b>3.57</b>     |

Pooling all seasons ( $\lambda = 3.858$ ) conceals that Summer has **4.7× the rate** of Spring. Any design based on the annual average is wrong for both seasons simultaneously — too conservative for spring, dangerously low for summer.

# Four Seasons, Four Distributions

The table gives four values of  $\hat{\lambda}$ . Those are four *different distributions* — not four adjustments to one.



# Predict — Rank the Seasons

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## 60 seconds · rank before you look

Tucson rain days, 2013–2022. Four seasonal  $\lambda$  values, in rain days per month:

**Rank these from wettest to driest — and estimate the ratio between the two extremes.**

- Summer (Jun–Aug)
- Fall (Sep–Nov)
- Winter (Dec–Feb)
- Spring (Mar–May)

**Then:** which is more likely — a **summer** month with zero rain days, or a **spring** month with zero rain days? By roughly how much?

## ✓ Summer 6.57 · Winter 3.90 · Fall 3.57 · Spring 1.40

| Season | $\lambda$ (rain days/month) | $P(X = 0)$ |
|--------|-----------------------------|------------|
| Summer | 6.5667                      | 0.141%     |
| Winter | 3.9000                      | 2.02%      |
| Fall   | 3.5667                      | 2.83%      |
| Spring | 1.4000                      | 24.66%     |

Summer is **4.7×** **wetter** than Spring by rate. But the zero-rain probabilities differ by a factor of **175**.

💡 About **one spring month in four** sees no rain at all. For summer it is about **one month in 700**. The rate ratio is 4.7×; the consequence ratio is 175× — because the exponential in the PMF amplifies whatever  $\lambda$  does.

# The Monsoon Signal: July vs. Spring

**July:** 103 rain days across 10 years  $\rightarrow \lambda_{\text{july}} = 103 / 10 = \mathbf{10.3 \text{ rain days per July}}$

**Spring month:**  $\lambda_{\text{spring}} = \mathbf{1.40 \text{ rain days per month}}$

Ratio:  $10.3 / 1.40 = \mathbf{7.4\times}$  — July produces 7 times as many rain events as a spring month.

| Question                 | Spring ( $\lambda = 1.40$ ) | July ( $\lambda = 10.3$ ) |
|--------------------------|-----------------------------|---------------------------|
| P(zero rain days)        | 24.66%                      | < 0.01%                   |
| P(more than 5 rain days) | 0.32%                       | 94.34%                    |
| Std deviation            | 1.18 days                   | 3.21 days                 |

A July with zero rain days is essentially impossible. A spring month with zero rain days occurs almost one in four months. The same Poisson model captures both — with different  $\lambda$ .

More than 5 rain days is a **1-in-300** event in spring and a near-certainty in July.

# Why Seasonal $\lambda$ Matters for Drainage Design

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**Wrong:** Design drain for the annual average (3.858 days/month).

**Right (ASHRAE 90.1):** Size for the peak monsoon month, not the annual average.

| Design basis                         | P(drain undersized in July)       |
|--------------------------------------|-----------------------------------|
| Annual average ( $\lambda = 3.858$ ) | High — ignores monsoon completely |
| Summer average ( $\lambda = 6.57$ )  | Lower — captures summer regime    |
| July rate ( $\lambda = 10.3$ )       | Lowest — sized for actual peak    |

**Principle:** condition before you model — use the  $\lambda$  for the operating context.

# Part 6

This Week's In-Class Exercise

# What You'll Do in the In-Class Exercise

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**Dataset:** `Week4_Tucson_Precipitation_Daily_2013_2022.csv` — 3,652 daily rows (2013–2022)

## Section A — Marginal Event Rate

- A1: Total rain days in the dataset
- A2: Overall  $\lambda$  (463 rain days / 120 months)
- A3–A5: PMF at  $k=0$ , PMF at  $k=5$ , CDF at  $k=5$  (overall  $\lambda$ )
- A6–A8: Summer rain days, Summer  $\lambda$ ,  $P(X=10 \mid \text{Summer})$

## Section B — Seasonal Poisson Analysis

- B1–B4: Spring rain days, Spring  $\lambda$ ,  $P(\text{dry spring month})$ ,  $P(\text{wet spring month})$
- B5–B8: July rain days, July  $\lambda$ ,  $P(\text{dry July})$ ,  $P(\text{extreme July} > 15 \text{ days})$

# Bridge to Machine Learning — Rare-Event Prediction

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| Week 4 tool                              | ML equivalent           | Connection                                      |
|------------------------------------------|-------------------------|-------------------------------------------------|
| $\lambda$ , the event rate               | Base rate / class prior | How often the event happens at all              |
| Poisson PMF over counts                  | Poisson regression      | Predicts a count, not a continuous value        |
| $P(X = 0)$ — the quiet month             | Class imbalance         | The 'nothing happened' class dominates the data |
| Seasonal $\lambda$ vs. overall $\lambda$ | Conditional base rate   | The rate depends on the regime you are in       |

A classifier does not output a label — it outputs a **probability**, the same kind of number `POISSON.DIST` returns. And when the event is rare, accuracy becomes a worthless score: a fault detector that always predicts "no fault" is right 98% of the time and catches nothing. Knowing the base rate  $\lambda$  is what tells you that.

# Three Deliverables This Week

|                   | What it covers                                | When                                       |
|-------------------|-----------------------------------------------|--------------------------------------------|
| Weekly Quiz       | 15 questions on this lecture and the Tutorial | Closes 8:00 AM Tuesday, Sep 22             |
| In-Class Exercise | Sections A and B, the charts, and the memo    | Work Wed + Fri · due Wed 9:00 AM next week |
| Homework          | A separate take-home problem set              | On your own · due Wed 9:00 AM next week    |

**The exercise spans two sessions.** You start it Wednesday after the tutorial, and finish it Friday. **You are not expected to finish in class** — use the session time for the parts where having me in the room actually helps, and complete the rest on your own.

💡 Three submissions, three D2L Assignments folders. The quiz closes 8:00 AM Tuesday, Sep 22; the exercise and the homework are both due Wednesday 9:00 AM — no late work is accepted.

## Looking Ahead — Week 5 Preview

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Your PMF table for Summer ( $\lambda = 6.57$ ) is bell-shaped, centered near  $k = 6-7$ .

Your PMF table for July ( $\lambda = 10.3$ ) is wider and centered near  $k = 10$ .

**The question to sit with:** What happens to the shape of a Poisson PMF as  $\lambda$  grows very large — say,  $\lambda = 50$  or  $\lambda = 100$ ? Which continuous distribution does it approach?

💡 Week 5 is the **Normal distribution**. At large  $\lambda$ ,  $\text{Poisson}(\lambda) \approx \text{Normal}(\mu=\lambda, \sigma^2=\lambda)$ . The discrete count distribution we built today converges to the continuous bell curve — and the tools generalize: instead of computing  $P(X=k)$  for integers, we will compute  $P(a \leq X \leq b)$  for any real interval.

# Summary — Week 4

Rare events have rates — and rates have distributions

Today's concepts:

- **Poisson model** — counts of independent events at constant rate  $\lambda$
- **PMF** —  $P(X = k) = e^{-\lambda} \lambda^k / k!$  · In Excel: `POISSON.DIST(k,  $\lambda$ , FALSE)`
- **CDF** —  $P(X \leq k)$  · In Excel: `POISSON.DIST(k,  $\lambda$ , TRUE)`
- **Exceedance** —  $P(X > k) = 1 - \text{CDF}(k)$
- **Return period** —  $P(\geq 1 \text{ event in } N \text{ years}) = 1 - e^{-(N/T)}$
- **$\lambda$  estimation** — count events, divide by periods; condition on season before modeling

**The core insight:** The mean ( $\lambda$ ) is necessary but not sufficient — the full PMF reveals the tail that the mean conceals.

## Key Formulas — PMF, CDF & Exceedance

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| Concept           | Formula                                  |
|-------------------|------------------------------------------|
| Poisson PMF       | $P(X = k) = e^{-\lambda} \lambda^k / k!$ |
| Mean and variance | $E[X] = \text{Var}[X] = \lambda$         |

## Key Formulas — Return Period & $\lambda$ Estimation

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| Concept             | Formula                                 |
|---------------------|-----------------------------------------|
| $\lambda$ from data | $\hat{\lambda} = \text{events/periods}$ |